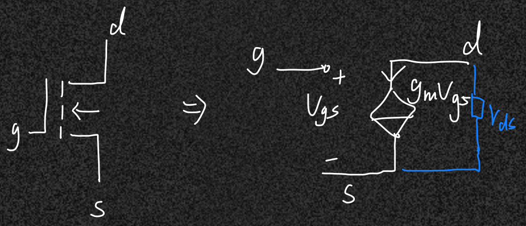
场效应管构成的放大电路

小信号模型

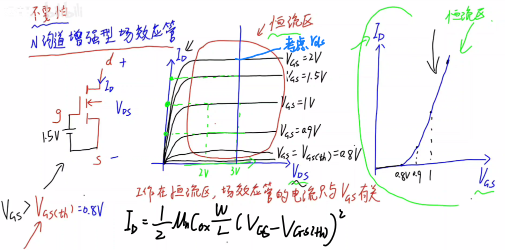


$$I_D = I_{DO}\left(\frac{V_{GS}}{V_{GS(th)}}-1\right)^2$$

$$g_m = \frac{\Delta I_D}{\Delta V_{gs}}$$

$$= \sqrt{I_{DQ}\cdot I_{DO}}\cdot\frac{2}{V_{GS(th)}}$$

$$= I_{DO}\cdot\left(\frac{V_{GS}}{V_{GS(th)}}-1\right)\cdot\frac{2}{V_{GS(th)}}$$



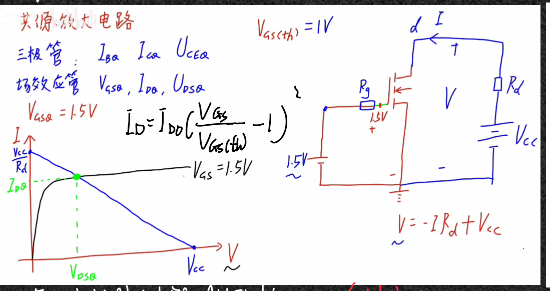
N沟道增强型场效应管

工作在恒流区，场效应管的电流只与V_{GS}有关

$V_{GS} > V_{GS(th)} = 0.8V$

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{GS(th)})^2$$

共源放大电路.



共源放大电路

三极管：I_{BQ} I_{CQ} U_{CEQ}

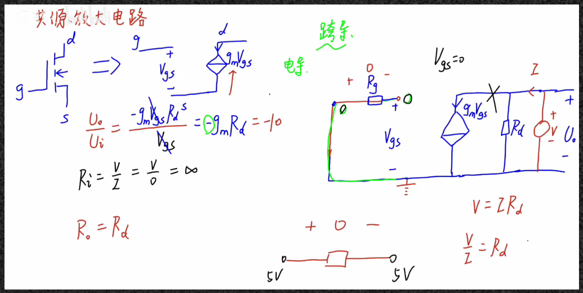
场效应管 V_{GSQ}, I_{DQ}, U_{DSQ}

$V_{GSQ} = 1.5V$

$V_{GS(th)} = 1V$

$$I_D = I_{DO}\left(\frac{V_{GS}}{V_{GS(th)}}-1\right)^2$$

$V = -IR_d + V_{CC}$



共源放大电路

$$\frac{U_o}{U_i} = \frac{-g_m\cancel{V_{gs}}R_d}{\cancel{V_{gs}}} = -g_mR_d = -10$$

$R_i = \frac{V}{I} = \frac{V}{0} = \infty$

$R_o = R_d$

跨导

电导

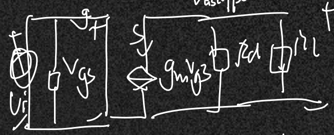
$V_{gs} = 0$

$V = IR_d$

$\frac{V}{I} = R_d$

共源的变种放大电路（结型、耗尽型）自给偏压

结型：$I_D = I_{DSS}\left(1-\frac{V_{GS}}{V_{GS(off)}}\right)^2$ $g_m = \frac{-2}{V_{GS(off)}}\sqrt{I_{DQ}\cdot I_{DSS}}$

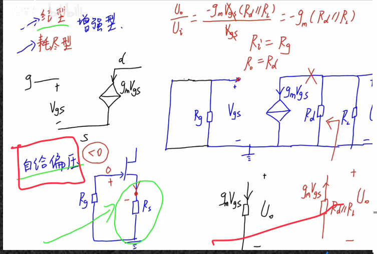


$$\left(\frac{1}{x}\right)' = \frac{-1}{x^2}$$

$$A_u = \frac{-g_m\cancel{V_{gs}}(R_d/\!/R_L)}{\cancel{V_{gs}}}$$

$R_i = R_g$

$R_o = R_d.$

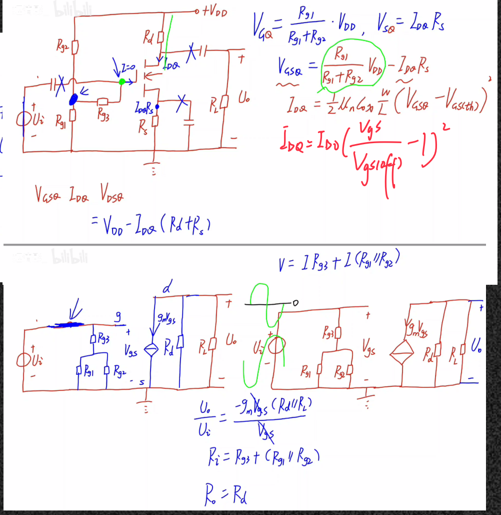


$$\frac{U_o}{U_i} = \frac{-g_m\cancel{V_{gs}}(R_d/\!/R_L)}{\cancel{V_{gs}}} = -g_m(R_d/\!/R_L)$$

$R_i = R_g$

$R_o = R_d$

同共射



$V_{GQ} = \frac{R_{g1}}{R_{g1}+R_{g2}}\cdot V_{DD}$, $V_{SQ} = I_{DQ}R_s$

$V_{GSQ} = \frac{R_{g1}}{R_{g1}+R_{g2}}V_{DD} - I_{DQ}R_s$

$I_{DQ} = \frac{1}{2}\mu_nC_{ox}\frac{W}{L}(V_{GSQ}-V_{GS(th)})^2$

$I_{DQ} = I_{DO}\left(\frac{V_{GS}}{V_{GS(off)}}-1\right)^2$

V_{GSQ} I_{DQ} V_{DSQ}

$= V_{DD} - I_{DQ}(R_d+R_s)$

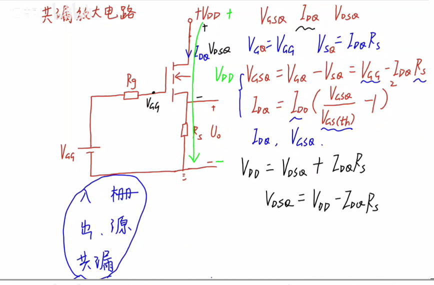
$V = IR_{g3} + I(R_{g1}/\!/R_{g2})$

$$\frac{U_o}{U_i} = \frac{-g_m\cancel{V_{gs}}(R_d/\!/R_L)}{\cancel{V_{gs}}}$$

$R_i = R_{g3} + (R_{g1}/\!/R_{g2})$

$R_o = R_d$

共漏放大电路.



共漏放大电路

V_{GSQ} I_{DQ} V_{DSQ}

$V_{GQ} = V_{GG}$ $V_{SQ} = I_{DQ}R_s$

$V_{GSQ} = V_{GQ} - V_{SQ} = V_{GG} - I_{DQ}R_s$

$I_{DQ} = I_{DO}\left(\frac{V_{GSQ}}{V_{GS(th)}}-1\right)^2$

I_{DQ}, V_{GSQ}.

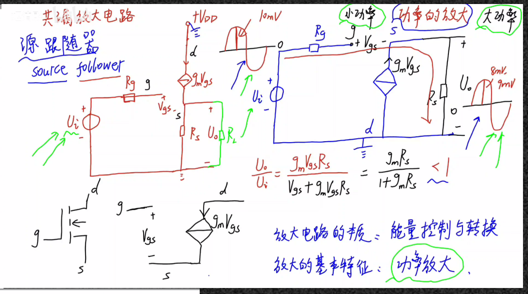
$V_{DD} = V_{DSQ} + I_{DQ}R_s$

$V_{DSQ} = V_{DD} - I_{DQ}R_s$

入 栅

出、源

共漏



共漏放大电路

源跟随器

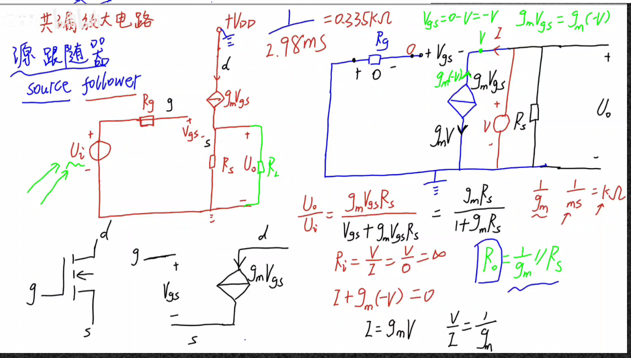
source follower

小功率 功率的放大 大功率

$$\frac{U_o}{U_i} = \frac{g_mV_{gs}R_s}{V_{gs}+g_mV_{gs}R_s} = \frac{g_mR_s}{1+g_mR_s} < 1$$

放大电路的本质：能量控制与转换

放大的基本特征：功率放大.



共漏放大电路

源跟随器

source follower

$\frac{1}{2.98mS} = 0.335k\Omega$

$V_{gs} = 0 - V = -V$ $g_mV_{gs} = g_m(-V)$

$$\frac{U_o}{U_i} = \frac{g_mV_{gs}R_s}{V_{gs}+g_mV_{gs}R_s} = \frac{g_mR_s}{1+g_mR_s}$$

$\frac{1}{g_m}$ $\frac{1}{mS} = k\Omega$

$R_i = \frac{V}{I} = \frac{V}{0} = \infty$

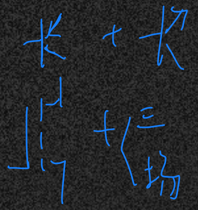
$\boxed{R_o} = \frac{1}{g_m}/\!/R_s$

$I + g_m(-V) = 0$

$I = g_mV$ $\frac{V}{I} = \frac{1}{g_m}$

同共集

复合管



: and so on

CONT
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